BO CHEAT SHEET LINUX & DEVOPS COT LOI

DevOps Personal Tutor System | Path: D:\Devops

1. CAU TRUC NGU PHAP CU PHAP LINUX (COMMAND LINE ANATOMY)

Cong thuc chung: Command [-Options/Flags] [Arguments]
• **Command:** File chuong trinh thuc thi (/bin, /usr/bin).
• **- (Co ngan 1 ky tu):** ls -la, rm -rf (Co the gop nhieu co).
• **-- (Co dai tu day du):** --help, --version, --no-pager.
• **> (Ghi de):** Xoa sach va ghi moi. | **>> (Ghi noi):** Ghi them vao cuoi file.
• **| (Pipe):** Dau noi dau ra (stdout) cua lenh trai lam dau vao (stdin) cho lenh phai.

2. CAY HE THONG FILE LINUX (FILESYSTEM HIERARCHY)

Thu muc	Khai niem & Vai tro cot loi
/	**Root Directory:** Thu muc goc toi cao cua he thong Linux.
/home	Thu muc ca nhan cua nguoi dung (vi du /home/minh123).
/etc	Noi chua toan bo **FILE CAU HINH** cua HDH va phan mem (Nginx, SSH...).
/var	Noi chua **DU LIEU BIEN DOI** (Log /var/log, cache, database).
/proc	Thu muc ao chua thong tin phan cung CPU/RAM & Process tu RAM.
/tmp	Thu muc chua file tam thoi (tu dong xoa khi reboot).
/mnt/d	Noi WSL2 mount o dia D: cua Windows vao Linux.

3. THAO TAC FILE & THU MUC CO BAN

```
pwd                                 # In duong dan thu muc hien tai
cd /duong/dan                       # Di chuyen thu muc (cd .. lui 1 cap, cd ~ ve home)
ls -la                              # In chi tiet tat ca file/folder ke ca file an .
mkdir -p folder1/folder2            # Tao chuoi thu muc long nhau cùng luc
touch filename.txt                  # Tao file rong moi
cp -r nguon dich                    # Sao chep file/folder (-r cho folder)
mv nguon dich                       # Di chuyen HOAC doi ten file/folder
rm -rf folder_name                  # Xoa de quy folder/file ep buoc khong hoi
cat filename                        # In toan bo noi dung file ra man hinh
head -n 10 filename                 # Xem 10 dong dau tien cua file
tail -n 20 filename                 # Xem 20 dong cuoi cung cua file
grep "tu_khoa" filename             # Loc tim kiem dong chua tu khoa
```

4. PHAN QUYEN & QUAN LY USER (chmod, chown, sudo)

Bang diem Octal: Read (r=4), Write (w=2), Execute (x=1). Doi tuong: User (u) | Group (g) | Others (o).

Ma so	Chuoi chu	Y nghia & Truong hop su dung
755	rwxr-xr-x	Mac dinh cho Thu muc & Script thuc thi (Owner toan quyen, Group/Others chi doc/chay).
644	rw-r--r--	Mac dinh cho File du lieu thuong (Owner doc/ghi, Group/Others chi doc).
600	rw-------	Bat buoc cho **SSH Private Key** (Chi Owner doc/ghi, cam tuyet doi nguoi khac).

```
chmod +x script.sh                  # Cap quyen thuc thi cho file script
chmod 600 ~/.ssh/id_ed25519         # Phan quyen chuan bao mat cho SSH Private Key
sudo chown root:root filename       # Doi nguoi so huu file sang user root
```

5. QUAN LY TIEN TRINH & DICH VU (PROCESS & SYSTEMD)

```
ps aux | grep ten_tien_trinh          # Tim PID (Process ID) cua tien trinh dang chay
kill -9                         # Ep buoc dung ngay lap tuc tien trinh bi treo
top / htop                           # Man hinh giam sat CPU/RAM realtime
sudo systemctl status nginx          # Xem trang thai dich vu Nginx
sudo systemctl start|stop|restart nginx # Khoi chay / Dung / Khoi dong lai dich vu
sudo systemctl enable nginx          # Cho phep dich vu tu bat khi khoi dong may
sudo journalctl -u nginx -n 50 --no-pager # Xem 50 dong log moi nhat cua Nginx
sudo nginx -t                        # Kiem tra cu phap file cau hinh Nginx
```

6. QUAN LY GOI & SUA SU CO (APT & INCIDENT RECOVERY)

```
sudo apt update && sudo apt install -y package_name # Cap nhat danh sach & cai dat goi
sudo apt purge -y package_name                      # Xoa sach goi + XOA TOAN BO CONFIG IN /et
c/
curl -I http://localhost                            # Kiem tra Header phan hoi HTTP Server
```

7. GIT & GITHUB CO BAN DANH CHO DEVOPS

```
ssh-keygen -t ed25519 -C "email@example.com" # Sinh SSH Key chuan Ed25519
ssh -T git@github.com                        # Kiem tra ket noi SSH toi GitHub
git init                                     # Khoi tao kho Git local
git branch -M main                           # Doi ten nhanh mac dinh thanh main
git status                                   # Kiem tra trang thai lam viec
git add .                                    # Dua tat ca file vao Staging Area
git commit -m "feat: mo ta cong viec"        # Chup anh luu tru commit
git remote add origin git@github.com:User/Repo.git # Ket noi remote GitHub
git remote -v                                # Kiem tra duong dan Remote hien co
git remote rename origion origin             # Sua loi go nham ten Remote
git push -u origin main                      # Day commit len nhanh main tren GitHub
```